

Extraneous Solutions and Real-World Applications

TEACHER KEY | Algebra II | Unit 2 | Day 6 | TEK A2.6.D, A2.6.E

Objective: I can identify extraneous solutions when solving absolute value equations, and solve the real-world equations from Day 4.

Vocabulary

Extraneous solution — a value the algebra produces that does **NOT** actually satisfy the **original** equation.

Why it happens here — splitting into two cases can create a solution where the "inside" expression has the **wrong sign** for that case.

Part 1 — A Solution That Does Not Check Out

Solve $3|x - 1| - 2 = 6x + 1$ and check BOTH answers in the original equation.

Isolate: $3|x - 1| = 6x + 3 \rightarrow |x - 1| = 2x + 1$

Split: $x - 1 = 2x + 1$ or $x - 1 = -(2x + 1)$

Solve: $x = -2$ or $x = 0$

Check $x = -2$: $3|-3| - 2 = 7$, $6(-2) + 1 = -11$. $7 \neq -11 \rightarrow$ **EXTRANEIOUS**

Check $x = 0$: $3|-1| - 2 = 1$, $6(0) + 1 = 1$. $1 = 1 \rightarrow$ **valid**

Solution set: $\{0\}$

Part 2 — Solve Day 4's Real-World Equations

Clarissa's houses: $|x - 1248| = 5$

$x = 1253$ or $x = 1243$ Both are even-numbered addresses within 5 houses, so **both are valid**

Part 3 — Surveying and Brighton Builders

Surveying: $1.5 = |x - 10|/10 \cdot 100$

$|x - 10| = 0.15$ $x = 10.15$ or $x = 9.85$ miles

Brighton Builders: $2|15x| + 52 = 292$

$|15x| = 120$ $x = 8$ or $x = -8$ (x is a suite count, so use $x = 8$)

That is 8 office suites per side per floor $\rightarrow$ **32** suites per floor $\rightarrow$ **608** total suites in the 20-story building

Practice

1. Solve $|x + 3| = 2x$ and check both solutions.

$x = 3$ or $x = -1$ (**extraneous**) Solution set: $\{3\}$

2. Solve $|2x - 1| = x + 4$ and check both solutions.

$x = 5$ or $x = -1$ Solution set: $\{-1, 5\}$ (both valid this time)

Exit Ticket

Solve $|x - 4| = 3x$, checking both solutions in the ORIGINAL equation. State the solution set.

$x = 1$ or $x = -2$ (**extraneous**) Solution set: $\{1\}$